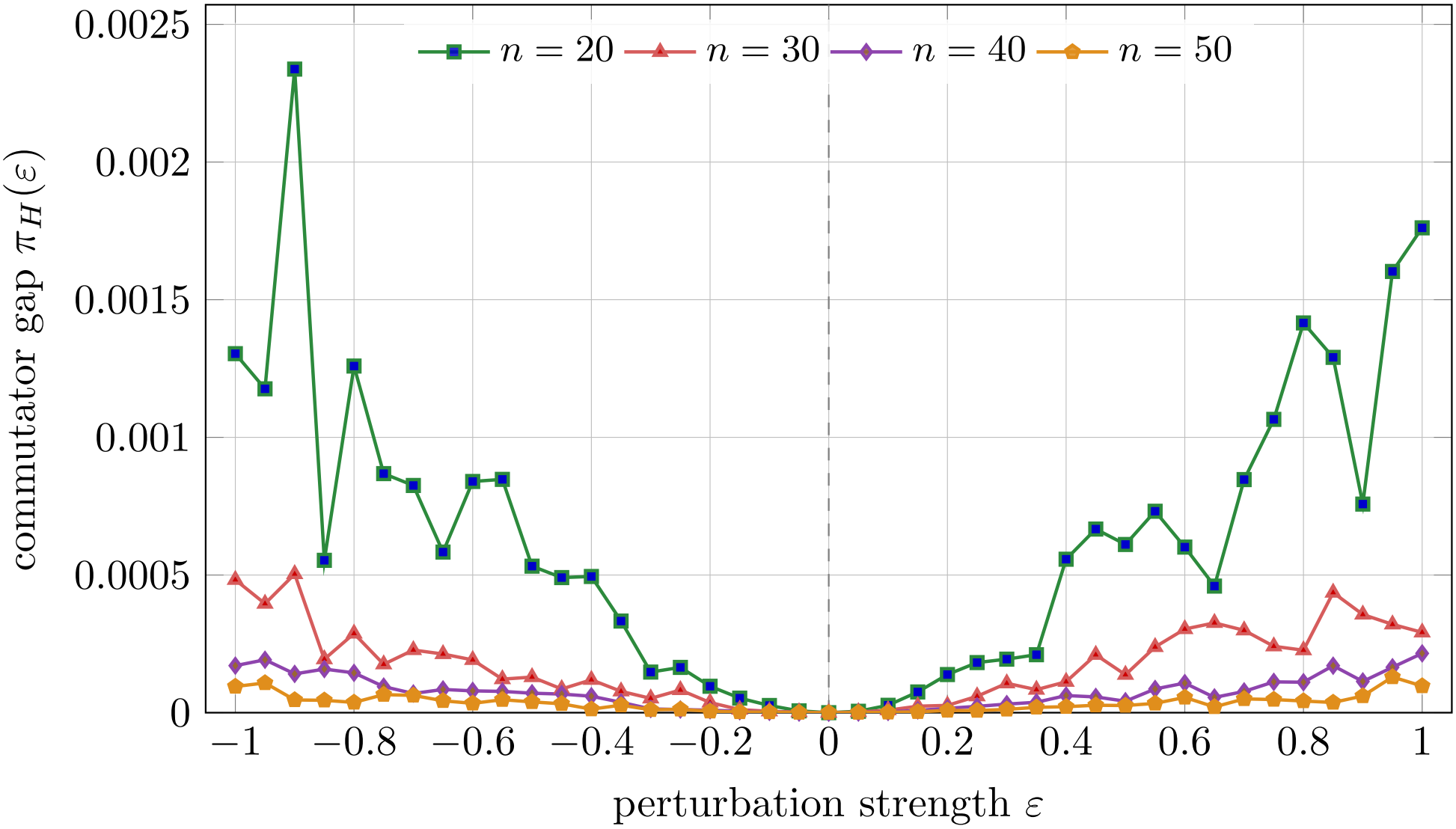

$H(\varepsilon) = \sum Z_i Z_{i+1} + \varepsilon \sum g_i X_i$, $g_i \sim \mathcal{N}(0,1)$, 30 samples per $\varepsilon \neq 0$
$n = 20$
$n = 30$
$n = 40$
$n = 50$
commutator gap $\pi_H(\varepsilon)$
0.0025
0.002
0.0015
0.001
0.0005
0
−1
−0.8
−0.6
−0.4
−0.2
0
0.2
0.4
0.6
0.8
1
perturbation strength ε